

ActInf Livestream #011.0 "Sophisticated Affective Inference Simulating An

Livestream | 1:35:41

all right

i think we are live hello everyone and

welcome

to the active inference live stream this

is active inference live stream

11 it is december 16 2020

and there's a lot to get to today so

thanks for tuning in

welcome to the active inference lab

everyone

we are an experiment in online team

communication

learning and practice related to active

inference

you can find us at our website

activeinference.org

on twitter as inference active at our

gmail account

our youtube channel or our public key

based team and username

this is a recorded and an archived live

stream

so please provide us feedback so we can

improve our work

whether during the live stream or after

all backgrounds and perspectives are

welcome here

and as far as video etiquette for live

streaming goes

mute if there's noise in your background

we'll always remember to use respectful

speech behavior

etc so thanks everyone just a few

announcements and points of process

there are two more meetings

for 2020 and they will both be from 7 30
to 9 00 a.m
pst and both of these
meetings are going to be about the
paper's sophisticated
affective inference simulating
anticipatory affective dynamics of
imagining future events a 2020 paper
from the iy
conference and that's what this 11.0 is
going to be primarily about
so if you're listening to this at the
end of 2020 then
uh we would love to have you at either
11.1
on december 22nd or 11.2 on december
29th
and also we are very happy to now
announce
for 2021
that's right that is going to be the
year 2021.
um we have announced our whole year's
schedule
so it's gonna be tuesdays from seven to
nine am
pst and all the instructions everything
you need to
know is at this link [rb.gy](https://rb.gy/kvnpyc)
kvn pyc and just to show you what that
link looks like
it is a spreadsheet and on the top
is information about how to participate
in the discussion
so fill out this form if you want to
participate email us with what dates you
want to attend
and then we will add you to the

discussions event that day
and then if you want to collaborate
check out this link which i'll get to in
a second
and just looking at a few of these
papers that we have up until the middle
of april
for every single paper we've been really
lucky enough and have an awesome
community of authors who want to join on
so we'll have an author representative
of every single paper
up through the middle of april and
beyond and we'll have
most weeks we'll have an author on so
we're going to get a lot of
opportunities to
hear from the authors about what they
were trying to do with their papers i'm
just really excited to hear from all
these different perspectives
and then of course by bringing on people
who are learning active inference or who
are from any different field
of expertise or experience we're going
to be able to
try out some new combinations that maybe
haven't been tried
speaking of the active inference lab
beyond the live stream
if you go to activeinference.org you'll
see
a call for collaboration for 2021
and this `aweber slash t slash ausm3`
link you'll be able to fill out a form
of interest and just let us know if you
might be interested in collaborating for
2021

because we have three main projects that we're really excited to be working on the first has to do with an active inference book of knowledge which is going to be like a participatory and an open and a very interesting designed textbook reference and uh source of online course material that'll be project one that's very um educational driven project two is gonna be the live stream and some other associated content types that we'll get to in 2021 and then project three is going to be tools for remote teams so this is going to be related to the frameworks for analysis that we laid out in our september 9th paper um viatkin at all and um we're going to be working on how we can do analysis and design for remote teams so that will be quite exciting all right thanks for all of that sorry about the relatively long introduction here we are in active stream 11.0 and the goal of acting stream 11 is to set the context for the next two weeks of discussion which are 11.1 and 11.2 the paper that we're going to be reading in both of these weeks is sophisticated affective inference simulating anticipatory affective dynamics of imagining future events

and that's by casper hesp at all 2020
and the researchgate link is provided
this video.0 is an introduction
or a context for some of these ideas
it's not a review
or a final word or a judgment the video
is just an attempt to contextualize some
of the ideas and the vocabulary as well
as some of the formalisms for the paper
which will be really helpful in
understanding the paper for people of
all backgrounds
the punchline of the paper is that we
can add features
to active inference such as anticipation
and affect and aiming for more powerful
simulated agents and perhaps new
insights for computational psychiatry so
we're going to be looking at it with two
main foci
one is on the simulating agents side
either computational agents or robots
the other side is sort of the human side
with
the computational psychiatry and how are
we going to be using this to have
insight into the human brain and mental
health
the sections of 11.0
first we will go through the keywords
and
that will be like a bridge or an on-ramp
for those who are from those different
fields
maybe learning about active inference
for the first time then we'll talk about
the aims and claims
abstract and the road map of the paper

how they get from a to z
then we'll go through the formalisms and
figures and in this paper's case
the formalisms are actually drawn
from a few other papers
so this video might go a little long and
it will contain information
on the two uh citations that
give us sophisticated affective
inference which is the sophisticated
side and the in
the uh affective site but we'll get to
there in a few slides
in eleven and uh point one and point two
we will discuss this paper so if you're
listening
in the live chat it'd be awesome if you
have any comments
uh as well as in the comments after the
video is done
or if you wanna put it in a twitter
thread or however just let us know what
questions
we could ask the authors and also the
panelists on
11.1 and 11.2 about this paper and if
those
discussions have already happened then
keep on adding comments because the
discussion never stops
all right so the keywords the best part
the keywords um as usual there's so much
to say
about each of these keywords and my
target here
is really the construction of the
on-ramp for this paper and the on-ramp
for active inference

so if you're in the active inference
community
i hope that these keywords are an
invitation to
explore some adjacent areas of research
and if you're just
learning about active inference and you
want to familiarize yourself more
then hopefully these keywords are like
an interface or a foundation
for your understanding and if you have
any thoughts or any
questions then always just leave a
comment or get in touch with us or come
on an active inference stream
because we're all really just learning
these ideas at this point and so
we just want to hear what people think
about
what are the connections that are
possible and
certainly my hasty production of these
videos would be greatly improved
by people in the live chat or comments
just
letting us know where we could take the
discussion
so the keywords of the paper included
anticipation
counter counterfactuals affect and
anxiety
so maybe some of these words are more
common than
others people talk about anticipating
events a lot in everyday life as well as
of course anxiety and affect or
emotionality
counterfactuals you don't hear as much

about in everyday life but
maybe one does they just don't use that
word so we'll come back to that in a
second
all right the first term is anticipation
so anticipation is a really cool term
and there's a lot of interesting
research
and my goal here is to really draw out
this thread of
anticipation itself apart from for
example
systems approaches to ecology or
cybernetic approaches to action
so we're going to really focus on this
anticipation link and that's going to
hopefully
enrich in all of these different areas
because we're going to see how they make
this puzzle that fits together
it's bigger than all the puzzle pieces
so we're going to start on the
anticipation side with a couple of key
citations
which are related to robert rosen who is
in the middle of the 1900s did some very
influential anticipation research
and then also this 2017 paper by louie
and some other recent review papers on
rosen's work
helped contextualize this discussion for
me on anticipation
so the axiom of anticipation as they say
is that life is anticipatory so we've
heard of life
as a replicator we've heard of life as a
meganotropic force locally like an
organizing force

and here we're going to have another facet and that's that life is anticipatory so we're really thinking about all kinds of systems just like we were thinking about definitions of life as being pattern making or replicating anticipating is now going to be one of our key features for what is life like at least according to this lineage of research and this is going to relate to a ton of other ideas including the counterfactual which is really important and the work of rosen and others on anticipatory systems led to them formalizing what an anticipatory system was and here was the actual diagram that they used and um again there's many perspectives on anticipation the cybernetics of russia and uh non-russian non-western anticipatory research but i'm just going to trace out this lineage of rosen for right now and certainly the concept isn't new but the systematic study of it was new when rosen was writing his book and so here's a quote from the um mathematical foundations that leads me to this figure it says rosen's rigorously mathematical study of this biology inspired subject led to a sequence of papers culminating in his book anticipatory

systems philosophical mathematical and
methodological foundations

so here's the definition that is
provided of an anticipatory system
an anticipatory system is a natural
system that contains an
internal predictive model of itself and
of its environment
which allows it to change state at an
instant

in accord with the model's predictions
pertaining to a later instant
an anticipatory system's present
behavior depends upon future states or
future inputs
generated by an internal predictive
model model-based behavior or more
specifically

anticipatory model-based behavior is the
essence of social economic and political
activity input

and that was drawn out as this m-e-s so
it looks kind of familiar to some
diagrams we've looked at in active
stream but it's actually different
letters representing different things
so m is the model of the system we might
say that's like the generative model
e is the effector those are like the
action states and then

s is going to be the system or the niche
and the environment includes the total
system and then the env the effectors
influence the system

and then there's a bi-directional
relationship with the model and the
effectors

now um i hope this isn't too far

down the rabbit hole but i think there's
a few things that active inference and
non-active inference people
can learn from this here because um
again

it helps to distinguish which parts of
what we're talking about are related to
for example inactivism or which parts
are related to

bayesian computation statistics which
parts are related to thermodynamics
and by tracing out the different
lineages of thought that culminate in
what we have today

we can also clarify not just uh previous
ideas and realign them in the context of
active inference

but also we can start to potentially
uh work on all the systems that were
within the scope of these disparate
ideas within the scope of what we have
and so it is very important to make the
mapping and to be specific about it
rather than just say well yeah active
inference reimagines it so it's going to
be totally different well it's not going
to be totally different actually because
a lot of these figures are from the
1970s

and they're going to remind a lot of
active inference people about probably a
lot of things

things that i'm not even perceiving
which is why i put it out there for
other active inference people and none
to be thinking about

so um we saw from an activism and from
the ecological psychological domains

that we're developing in the 1900s
that there's many rich ways to think
about this
shared interaction or the interface
between the ecological niche and the
skilled and intentional organism
we also saw from multi-scale systems
complex systems that there's many ways
to think about internalism and
externalism
and about causal relationships there's
no privileged level of causation
or other multi-scale system approaches
okay
now we're going to focus on anticipation
not these other
multi-scale domains we're going to be
one domain one type of niche
what are we going to really be looking
at with anticipation
okay so um the
uh anticipation research frames this
world
as a multi-loop situation there are
simple or complex versions
that include the formers the observer's
formal model
f as well as the natural system so down
here on the bottom
we have f which is the formal system of
the
agent that's like internal to the agent
and then
we have n the actual external natural
system and so we've seen this again in
the context of internalism and
externalism and in action and inference
but here it's actually only about

anticipation that's why i think it's so interesting especially because it's actually called inference what the agent is doing with the formal model and that's inference under a formal model so a lot of um hopefully connections are being made and the idea of the anticipation research is that we can find analogies amongst natural systems within rosen's holistic anticipatory ecology so one formal system this is going to be our formal ecology and we're going to go out to a bunch of natural systems deserts and rainforests and we're going to be doing a decoding process and by decoding the causal entailments of the natural system by inferring what kind of feedback loops are uh anticipating and being anticipated by that system which is our own loopyness over here by undergoing that process we'll be able to find structural relationships between different natural systems so here's natural system one and natural system two and this line of research is interesting because it introduces a basis in logical areas as well as in category theory and dynamical systems so it bridges a lot of areas of formalisms and that enables this kind of law-like and semantic or ontological addressment

of systems which are things that the
active inference
research frontier is also focused on
like um
is active inference a theory of
semantics and so
um i think there's a couple ways in
which
active inference could come to this
anticipatory framework and have some
insights
that are evaluated from active inference
so
first the formal model of the organism
doesn't have to be formal in any sense
that we traditionally understand
but it is something that is formalizable
or maybe it's related to the church
turing hypothesis
or there's a lot of ways to think about
this but active inference is pretty
vague on purpose about the exact nature
of this model by just saying
well it's something that we're also
inferring as an observer who can make
a certain kind of model so we're not
sure that we're on the right one
but this f is a generative model so it
doesn't have to be just an analytical or
inferential model
it's actually related to control systems
so here it's encoding and decoding
but we talked a lot about action and
about perception
so active inference more squarely places
this loop
not in the realm of signal and
abstraction and meaning like semiosis

but within the control theory loop of
system state
and internal states and effectors
um second the formal model as we've
talked about with the
internalism and externalism and tail of
two densities the internal model or the
formal model
doesn't have to be simply inside of the
organism
it can exist in this exocortex way in
this extended or embedded
and cultured or a collective way and so
it doesn't need to be all within the
bounds of the organism now rosen
probably was aware of all these tensions
and used
fuzzy sets and ecological thinking a lot
of other things so it's not again
i'm not really commenting on the
anticipation sorry anticipation research
don't know too much about it but i'm
just thinking about where we could draw
and build upon and
really this is so interesting that in
the anticipation
area there's a dialectic or there's a
dynamic between the formal system and
the natural system so it's almost like
natural and artificial
and then that's related to encoding and
decoding so it's framed about
information and about
natural communications with an unnatural
almost
whereas the active inference approach
takes that ecological insight
and connects it to control theory

cybernetics so it's action oriented
that's the action research
side as well instead of the semiotic
elements and seeing the semi-symbiosis
as emerging
out of action rather than seeing the
semiosis as primary
and stimulating or signaling action
which one is primary well
which one was around first and uh
again really important precursor
research for complex systems a lot of
other areas
and just we're coming at it from
anticipation
we're not using too much active
inference jargon we're just thinking
wow if active inference could map to
anticipation then every keyword in
literature about anticipation
we could be able to be one jump away so
that's why semantic indexing of research
databases is so important as well
all right second key word
counterfactuals so counterfactuals
are very related to prediction and to
anticipation
and that is why um they're in this order
so many philosophers and this is drawing
from the stanford encyclopedia
philosophy
many philosophers have proposed uh to
analyze causal concepts in terms of
counterfactuals
and the basic idea is that a causes c
and that's related to this claim that if
a had not occurred
c would not have occurred so that's

called the difference-making cause or
the cause that makes a difference
so it's like you imagine two worlds
where something happens or it doesn't
happen
and then you ask about how they'd be
different so if tomorrow
there was no sunrise in california then
it wouldn't be the temperature it is
today
so you can think about an alternate
world even one that can't happen or
wouldn't happen
and then you can talk about that and say
oh well that's because the sun causes
the day to get warmer when it rises or
when the earth rises should i say
and what's good about this
counterfactual is that you can go beyond
simple counterfactuals and you can
actually have quite nuanced logic
of counterfactuals so here was an
example from lewis 73
where depending on the way that more
information is added
uh it changes the truth value of the
counterfactual so
here's just an example of how that can
flip and then also how we can use logic
not going too deep but just we can use
logic to denote these sentences
so if i had shirked my duty like not um
signed up
when i had to no harm would have been
sued no harm would have followed so
if there would have been i this event i
then
not h no harm and then in b

we see if i had shirt my duty and you
had two so now it's kind of like the and
operator i
and u two events happen event i and
event u then
there would have been h and then here in
part c
we see i and u and part t
then not h and so
this is kind of like a formalization but
it is can also be done
in ways that are a little bit more data
driven
instead of formal framework driven but
it's very related to
so many things that we've talked about
like infinite grammars and
logical sequences and about the ways
that you can have a generative model
and then you can ask about how things
are based upon how things
could be in a different sense we're
going to of course come back to that
with the papers
here is where the counterfactuals come a
little bit more directly to inference
so counterfactuals in the inference case
are like can we understand how things
are or were by asking how things
could be that's purely about inference
and then the action
uh orientation of that question is can
we reduce uncertainty about future
states like can i make sure that i
have a good core body temperature by
asking about how things could be
so that i can choose the right actions
so if it gets cold later tonight

and i'm going to want to have to have a
jacket or feel warm then i might need to
get a jacket right now
and so that's what this paper is about
that's what we're going to return to in
precision
but let's keep it within the sort of
keyword level analysis
and here's a plot from uh scp from
stanford encyclopedia philosophy and
it's a strict
analysis so categorical of how the
antecedent what happens before
not the ant the antecedent is related to
what does
or does not that's what this kind of um
I means not
the consequence the consequences are
what happened after so
in this situation we have things that
actually happened
before above the above yeah the x-axis
and then things that didn't happen
before like you know if
the sun wouldn't have risen yesterday
then that's this not
phi that's not what happened and then
the consequences are
shown on this symmetry axis so in this
quadrant on the top left we have things
that
did happen before and they did follow so
this is like today
i tied my shoe and it did stay on so
this is sort of normal reality
the shaded region must be empty why
because
it's things that actually did happen in

the antecedent but didn't happen
in the uh consequent so this is like
things that where you know
i boiled the water but then it wasn't
hot or something like that like
something where
you did the setup correctly but then it
didn't occur so we know that that
doesn't really
happen and then there's this whole set
of
worlds below the x-axis which is like if
anything had been different in the past
if i had been wearing a different color
shirt if dot
dot dot all of those worlds whether
what actually happens happened or not
all of the worlds of the past
are accessible because they might relate
to a similar world outcome or not
and we can nuance this sort of
categorical approach
to what's called a similarity analysis
and so a similarity analysis
is where we can start with our estimate
of our current state
current moment the causal relationships
and all that sort of stuff at this
origin point and then with the same axes
of what did or didn't actually lead up
to the event
and then what did or actually didn't
follow from the event
we can have more of a continuum so
um yes there's still little shaded area
here but
as you get to events that are further
and further in the future

you get to things that potentially
there's more latitude
but the sphere is a similarity idea it's
kind of like philosophers discovering
numbers they're going from the
categories and they're like maybe it's a
one through ten scale
maybe it's a continuum maybe we could
take an integral over it
maybe we could take a derivative what
would all these things mean and so
that's kind of the
fruitful discussion with people who have
a quantitative and a qualitative
background
because the rigor of this approach which
is
barely scraped out with this just
summary abstract that i have
it's a lot of horsepower to plug into if
we can reconsider
some of these topics in terms of a more
integrative way of thinking
then we can come back to everything
about antecedent and consequent
and reinterpret it and re-enliven it so
that's what's really exciting about
transdisciplinary research and teams
hopefully not to
always be returning to that point though
but it's really true
all right affect so there's many usages
of the word affect just looking at the
dictionary definition
uh there was just so many and
i i found this image and i thought i
don't even know what to say about this
image

because person a has an angry emotional
affect

so that's like using their character
about them as their they had an affected
character

and then trying to hide their anger
person a put on

an affect because affect has a
uh connotation of being how other people
perceive your manner of speech and
affect also has a sense that means
affected like changed altered in a way
once he entered the room

he started speaking in a fact in an
affected manner

now doesn't really describe anything it
just says he did it in a different
manner but

still person a then affects affected
that's what this is trying to show in
this left cartoon i believe affected
person b by pushing them into the pool
the

effect of this a versus e
was that person b ended up in the pool
right side

so that's what i think was trying to be
shown was the common sense and the
confusion which is the
affect is the cause and the effect is
the consequence

so that's why it's interesting because
affect and effect
of course common typo but it relates to
what we just saw
about the uh precedent the antecedent
and the consequent
in terms of counterfactual structures

the world but also
affect has to do with of course
emotionality that's why we're here
that's what interests us
and so in that more classical
psychological sense
affect is considered to be related to
like emotion or experience or
qualia again not my area just from my
skimming in preparation for this video
and often
affect is displayed on these lower
dimension manifolds or axes of variation
so for example one axis might be from
relaxed to stimulated now that doesn't
mean feeling good or bad you can be
feeling good and relaxed or bad or tired
or
um here there's a mild too intense
so these are emotions that don't grab
your salience so much
and these are ones at the high intensity
that do like being astonished or being
odd or shocked
then there is a orthogonal so
at a independent angle axis on the x
here
that goes from unpleasant to pleasant
and so whether you think of it as a two
by two like there's the pleasant
and intense experiences in the top right
and mild and unpleasant down here being
a little bit too cold or something like
that
you can think of it in a two by two you
can think of it in a circle
you could think of it as having more
dimensions you can think of it as

having measurements in this phase space
so if this is
maybe the first time you've seen all the
emotions on a chart think about what all
the options of charts
are and then all the options of emotions
and then that's kind of what we're going
to be
getting at so not just points in space
but we could actually be talking about
trajectories
through that phase space whether it's an
emotional cascade or
a pathway to radicalization like we were
talking about
yesterday in um 10.2 whatever it happens
to be or just your
daily routine if you wake up and you're
here how
are we going to get to here this is from
feeling afraid
through feeling excited and happy into
content and serene
now not that this is a pathway or the
pathway or whatever
but perhaps if we use tools like active
inference or other computational
frameworks
could it be possible to introduce some
ways of thinking about this diagram that
would help us
actually navigate these ridges and find
paths find policies
for people who are in one area of this
face space
to find a trajectory into different
parts of that phase space
so they already have the phase model of

the valence and the effect
and then there's already this trajectory
model the fields and trajectories coming
together
and now we're gonna reconsider
and um here's just a couple citations on
the left about
non-active inference neuroscience
approaches to
valence and to affect
valence being positive to negative just
like an electron or a proton has like a
valence
um valence is positive to negative axis
so that's the x-axis here
it's considered to be one of the major
axes
of variation as people who feel good or
bad
or change how they feel throughout time
can attest to
and so again we want to think about
affect in a couple senses not just the
semantic
we want to think about affect as a
statistical uh
phenomena especially one that can be
studied in all kinds of animals as well
as in
even computational agents and in that
situation
affect as a statistical phenomena is
like
feeling greedy or feeling volatile
or feeling regretful or feeling
confident
these are terms that we might want to
apply to even a chess playing

statistical machine or a gambling
machine oh it's really
acting shy the intentional stance from
dennett
maybe that's a valid um thing to say but
the statistical sense of affect is just
purely the way that the system's
behaving
how would we describe its intentionality
as if
that's very behaviorist um very from the
outside
then there's the neuro side of affect
and that's related to the neuroscience
kind of literature that's here people
who are actually looking in the brain
for
electro and chemical differences and
people who have different affect and
different affect states connecting it to
different features
uh dynamical or otherwise of the brain
so that approach is very internalist
it's focused on things internal to the
organism
and it tends towards this sort of neuro
scientific reductionism
and then there's also affect in the
psychological and in the psychiatric
sense which is the experience and that's
where of course this whole
area this field of emotions that maybe
some people experience a little
differently than others but
no one strongly disagrees or says that
it's preferable to go on
this side or the other i mean that's
kind of the question in many ways of

cognitive diversity
though and this is where i believe
active inference comes into play
is you have the statistical behaviorist
could be about robots could be about
anything
you have the neuroscientific very
internalist very
neuro human related and then you have
the psychological psychiatric which is
very social
and very semantic and based upon the
clinical relationship in many cases
here's where active inference comes into
play it adds in
a level of computational uh
computability
but also philosophical agnosticism about
a system
that using systems engineering
other approaches we're going to be able
to talk about systems very precisely
while also remaining basically agnostic
on
several key issues so when people
think that some aspect of uncertainty
in a framework is a simply negative
thing it's not that i'm doing
apologetics here i don't think
every uncertainty is a good thing
certainly not but there are several
areas just like we discussed at the end
of 10.2
we're actually saying yeah we're using
it as a tool so
whether the person chooses to have
cultural script a or cultural script b
about this experience we're not going to

make a value judgment on that
but potentially from a purely
statistical
framework we just want to be able to
describe certain kinds of behavioral
state transition matrices being
different in our model
so that's a way where people can have a
cultural experience in a personal
experience
but also use tools that are empowering
so um also beyond the computational and
the systems tractability we bring in
this whole control
and action perspective because all these
are about thoughts
you know words words words thoughts
thoughts thoughts
and rumination and overthinking and
underthinking can also be
a maladaptive state for individuals or
for collectives
so how are we going to orient this
psychiatry especially in action instead
of just like
you're in brain state a so now we need
to zap you or give you a chemical to put
you into brain state b
well it's not so simple or at least we
haven't figured out the simple way to do
it
yet so how are we going to think
big about the niche and the culture and
the extended environment
and then ask about how can we navigate
this
um you know care support team
experienter network how can we navigate

through this landscape
which is defined by specifics for this
person in their situation
how are we going to find a productive
policy to get from here to here
so those are the kinds of things that
get me really excited
because i feel like they're strong yet
neutral
and um powerful ways to talk about
mental states and to give one example
of a mental state which was a key word
so i have to go into it for this section
but um it's something that's you know in
our experience is anxiety
so i'm not a medical doctor of course
and i'm not your medical doctor so this
is really about that computational
psychiatry
perspective though um we all
feel anxious we all want to help each
other
no one is alone or should feel alone so
we all want to work together
and help get you on a productive path
wherever you're at
all right so here's the definition of
anxiety that's given by none other than
anxiety.org
anxiety is the mind and body's reaction
to stressful
dangerous or unfamiliar situations it is
the
sense of uneasiness distress or dread
you feel before a significant event
a certain level of anxiety helps us stay
alert and aware
but for those suffering from an anxiety

disorder it feels far from normal it can be completely debilitating and in the anxiety.org literature they provide three different categories of anxiety disorders again this is contextualizing why we're studying this area as well as trying to find what are the footholds where active inference could start to make an impact whether in people's lives or in the biomedical system or in the health insurance system just where across the board are we going to see this insight take place in systems change so the three kinds of categories of anxiety that they talk about are uh one anxiety disorders which are general featured by excessive fear and anxiety kind of i don't it's a little self-explanatory but uh second category is obsessive-compulsive and related disorders which are related to obsessive intrusive thoughts and compulsive behaviors these behaviors are performed to alleviate anxiety associated with obsessive thoughts in their quote and then the third category is trauma and stressor related so one time or chronic stresses and all these different interacting factors that are so

nuanced that is related to this third category

so what to say about anxiety i mean wow

so first is there's just many kinds and there's many experiences

so we want to respect that everyone's having their own experience

and it's going to be a totally unique experience to them

and it will demand a personalized way to help them just like everyone so that was sort of the first

point then there's of course to say that there's many aspects to this and there's many causes and factors

just from the clustering of diagnoses that can be considered within the category of anxiety disorders

according to the dsm let's say it's quite broad there's many

uh features that some people might present with versus others and there's no clarity on the underlying neurophysiological or even situational basis

and also it influences many kinds of thoughts and behaviors

but what's a theme in anxiety and something that connects

all these different areas is uncertainty is

a theme and when we see the word uncertainty in active inference

it's like a doorway because active inference is all about

agents that are trying to reduce their own uncertainty

about action in their niche so

we see uncertainty and anxieties about
uncertainty
we want to think about how we could
build active inference into a broader
discussion
about psychology and psychiatry so let's
think about these
different manifestations of anxiety and
reframe them in the context of active
inference and a couple of other
topics that's partially where this paper
is coming and we can start um
with the uh integration of all the
keywords together before we go into the
paper
so starting with anxiety let's go first
to
affect so anxiety it's either a
dimension
of affect or it's a point in affect but
the point is
it's someone's lived experience that
they're having anxiety
and specifically that they're feeling
negative about it and they want to get
better
because if it's not feeling negative
about it then it's not
pathological that would be considered i
believe within the you know
realm of healthy but not suffering but
if the person says puts their hands up
and says yes i'm suffering i would like
to have less anxiety
that's what we're talking about in this
situation
as well as managing it for everyone
whether they realize that at a

metacognitive level or not
the next keyword is anticipation
and anxiety can be very related to
anticipation
which is like predictions or estimates
for the future
and here's one thing that we can unpack
at this point as we start to turn
towards more mathematical sections for
the rest of the talk
and it's actually that the estimate is
really like a package of a couple of
things
it's the specific contents of the
expectation
but also it's its likelihood
relative to other things that could
happen and its confidence
so that's like a way to say that we can
separate what the prediction is from how
likely it is
and from how likely we think our
prediction is going to be good
and let's combine that with a few
examples with the counterfactuals
okay because i said it's how likely
something is relative to other outcomes
and that's related to the counter
factuals that's related to those spheres
of similarities and those similar
worlds and so counterfactuals are
essential for anticipation
but they're also critical in the current
moment so let's think about
anxiety as a theme and counter factuals
and anticipation
and about the past present future so a
counter factual about the past

that might lead to anxiety would be like

i wish that in the past i had done x

instead of z

or i wish i never did z you know two

hours ago

or 20 hours ago or 20 years ago

and then a counterfactual about the

current moment is like i wish it were

like

x right now instead of like z right now

just i wish something were different in

a state

about the world but here's where we get

those confidence estimators and the

anticipatory systems element because we

don't have to anticipate the past or the

present that's just kind of how times

arrow is funny huh

but when we're talking about the future

people don't really include these

confidence estimators

about the present they don't say i'm 100

confident that it's daytime

they say something like it just is

daytime

but when they include something about

the future it's very common to include a

confidence estimator like

i think it's unlikely that my friend

will do x in the next three years

and then you can have always predictions

amidst other predictions so like i think

there's a 10 chance that my baseball

team wins the world series

that means that there's a 90 chance that

you don't think they're going to win the

world series

and then you could have a critique or a

metacognitive estimate for your estimate

like yeah i said 10 and 90. but actually

i have no idea

so it could be way way way off so you

have a very loose estimate even on the

10 and 90

other times you're like yes i'm very

very confident now again you could be

wrong

you could be way off you could be asking

the wrong questions

you could be you know not saving the

text file

so the world intervenes in ways that go

beyond this model

but we're thinking about just this

simple decision-making context

and so in the context of anxiety

we can think about the generative model

that leads to

the production of affect of experience

whether it's epiphenomenal or not

later day but this generative model is

in this strange attractor of a zone of

thought and action

and that strange attractor is whether

it's ruminating or whether

it's related to any of these other

manifestations of anxiety

it's like there's a strange attractor of

thought and then

that can be overwhelming or it can lead

to negative experiences or it can lead

through a trajectory that goes to

negative places

so we want to be maybe moving towards

thinking about this landscape of

attractors of thought and action and

niche
and social environment thinking about
all these factors together and then this
field of affordances
the field of consciousness the free
energy landscape
how can we bring all these things
together think about how to help people
who are experiencing anxiety
and other situations and
before we go to the paper it's this
slide we'll return to
and it's what does free energy principle
and active inference say about the
relationship between agents in the world
when in this case the agent is affective
emotional
and imaginative and here's the affective
agent
who's also imagining future emotions so
this is the kind of model that we want
to keep in mind for the paper
it's about an agent who's acting in the
world um
and right now it's just a simple
interface but you can imagine more
nuanced interfaces at a later model
and this agent is having an experience
but that's not um a philosophical claim
it's just a parameter that it's
calculating
and then it is also calculating
perceived
future likelihoods of different states
okay so with no technical details at all
that's what this paper is about agent in
the world
with this valence affect here it's

simple it's zero to one like good to bad

but

you could go in different dimensions but

here it's just

good to bad agent in the world thinking

about the future

okay now it's going to get more

technical

that's what the paper is for though so

sophisticated affect of inference

simulating anticipatory affective

dynamics of imagining future events

it's from october 2020 so just like two

months ago

um at the first international workshop

on active inference iy 2020

at ghent and the authors are

listed here so cool first annual

good bold start we wish them the best of

luck in their

series the aims and the claims of the

paper are as follows

so they write in this paper we aim to

provide a mechanistic account of how

afflictive responses can be generated by

imagined future outcomes and how this

can become

dysfunctional during rumination by

combining

two recent developments in active

inference anticipation and affect

we provide a formal model of these

phenomena and simulate how overthinking

a situation can occur

continuing to the point where unlikely

yet aversive and arousing situations

emerge in one's

imagination so interesting motivation

for the paper

no and in non-active inference

terms i would frame this as how can we
model

affective and anticipatory active

inference agents

what features do these simulated agents

display in simple environments

and then because this is such a short

and inter introductory paper

how could we think about this being

useful in the fields of psychiatry

cybernetics all these different areas

and of course we'll go into the

formalisms soon

so the abstract of the paper in this

paper we combine sophisticated and deep
parametric

active inference to create an agent

whose affective states change as a

consequence of its bayesian beliefs

about how

possible future outcomes will affect

future beliefs

to achieve this we augment markov

decision processes with a bayes adaptive

deep temporal tree search that is guided

by a free energy functional

which recursively scores counterfactual

futures so we're already seeing a lot of

the key words come back into play

our model reproduces the common

phenomena of rumination

over a situation until unlikely yet

aversive and arousing situations emerge

in one's imagination

as a proof of concept we show how

certain hyper parameters give rise to

neurocognitive dynamics that
characterize imagination-induced anxiety
and so we're going to use the
computational psychiatry definition of
anxiety
and uh yep it's a short abstract
relatively short paper
invokes a lot of formalisms and so i
hope that people find it interesting
the roadmap is pretty short we only have
to go to a few gas stations on this one
there's an introduction there's a
methods
figure one is a directed acyclic
bayesian graph
which we'll build up to figure two is an
illustration of the state space of the
task with all four states
and then table three has a lot of
formalism and it's the predictive
posteriors that provide
the empirical priors for the generative
model
three is the results and then there's
figure three which is just
an example of simulation results showing
detrimental effects of overthinking
so they uh made it about overthinking
but they didn't overthink the complexity
of the paper
so it's a simple straightforward paper
but again it invokes a lot so
let's get to the formalisms and let's
try to really understand them or at
least
give them a space where they can exist
even if we don't know all the details
where are the edges of the formalism

so that we could be thinking about it
correctly or asking the right questions
or just the questions that make sense
for us
whatever they are that's the best
question to ask best question to ask in
a live chat
best question to ask in a comment or on
a live stream but
definitely just ask it because someone
else is going to be asking it and you'll
be really helping them
so formalisms here's from the paper
they wrote by combining the
ensuing recursive update scheme of
sophisticated inference
with deep parametric affective inference
i don't have the british accent to say
it the right way though
we can derive a general purpose
generative model of the following
mathematical form summarized graphically
in figure one and
in tabular form in table one okay so
they're combining two citations here
and so it's worthwhile to ask what these
citations were
and they are sophisticated inference
that's first in at all 2020 the first
citation in the sentence
and then there is hesp at all's paper
deeply felt affect which was actually
published november 30th
2020 so it was a pre-print i think at
the time of this one
being written but then now it's been
published
so here's how to think about this

paper sophisticated affective inference
that's what everyone is listening to
this video about that's what this paper
is about that's what the discussion is
mostly about
so it's combining you know building on
the shoulders of other
shoulders the affect side
with the temporal depth side both of
which we're going to go into
in this video which is why it's a
background video
this paper builds on top of them another
brick in the wall
another uh citation in the network and
we're going to be thinking about where
does that go given that this was in
october 2020 and it's only december 2020
um or maybe even only a few months later
if you're listening to this but where
are the next places to go
well there are the directions of making
more advanced simulations
like taking something that they took as
a fixed variable
in this paper and just asking if it
could be a vector of variables or if it
could be a continuum
or if it could be another type of model
inside of that and then
give it a new acronym and make it
another meme so
what simulation directions could we go
just purely exploring or making it
more mappable to other fields then um
how could we move in a way to integrate
better with
computational psychiatry and psychology

so how can we integrate with experts and
with vocabulary and
key questions in those fields so that
this isn't just like a machine
learning parallel universe related to
mental health
but it's actually something that's
connected to how
social workers in the field or how
caretakers in their
daily experience relate to these topics
and then another direction which i don't
know
too much about but i hope that we can
have some people on to
teach us more um are about new models
for robotics
so how do does including these kind of
top level parameters like affect and
in the context of a deeply uh nuanced
counterfactual anticipatory model
how does that relate to a robotics
context and will that help you know
robotic birds finally come out for
public use
well let's go to sophisticated inference
first so again
temporal depth and affect are the two
pillars that we're going to build on
we're going to go depth first search
some of you are going to get that here
is the sophisticated inference paper
which we're not going to read for an
active stream
so i will just read the abstract
active inference offers the first
principal account of sentient behavior
from which special and important cases

can be derived

for example reinforcement learning

active learning bayes optimal inference

bayes optimal design etc

active inference resolves the

exploration exploitation dilemma in

relation to prior preferences by placing

information gain

on the same footing as reward or value

in brief

octave inference replaces value

functions with functionals

of bayesian beliefs in the form of an

expected variational free energy

key point in this paper we consider a

sophisticated kind of active inference

using a recursive form of expected

free energy sophistication describes the

degree to which an agent has beliefs

about beliefs

so higher order theories

we consider agents with beliefs about

the counterfactual consequences of

action for states of affairs and beliefs

about those latent states

in other words we move from simply

considering beliefs about

what would happen if i did that so

that's a first degree counterfactual

to what would i believe about what would

happen if i did that that's a second

degree counterfactual

so for example it could be i really want

to know about

a pen's relationship to my affordance of

writing but i don't know how heavy it is

and so you can keep on thinking about

well what should i do with my hand

to find out how heavy it is so i can
resolve my uncertainty about it being in
affordance
these are the kinds of nested and
open-ended frameworks
for cause which can be extremely
contextual and ecologically
conditioned we want to set up these
causal chains in a ways that that we can
capture them
that's what this is about but the state
space is big
so a common theme we're going to come
back to is like how do we look through
these big
state spaces the recursive form of the
free energy functional effectively
implements a deep tree search
over actions and outcomes in the future
crucially
this search is over sequences of belief
states as opposed to states per se
we illustrate the competence of this
scheme using numerical simulations of
deep decision problems
so um one way of interpreting this last
part about here is this search is over
sequences of belief states opposed to
states per se
this is really taking that full jump
into the world of action instead of just
inference on world states
so instead of like i'm going to have all
my computational power
on what the temperature is going to be
in california in one year from today
it could be like i should just have
between one and three jackets

and then within an 80 20 but really more like 99

1 all have the right policy

but no one puts the big computational power on the jacket prediction algorithm they go on the climate prediction algorithm and there's a lot to say about that

why that decision is made not in that specific case but just in general why do we go with all of our power

on predicting states in the world rather than doing

prediction on policy when in the end policy is like what we can control and bound and then we can leave space for the unknown we can leave space for how likely is this tsunami every once in 30 years or 50 or 100

what's the most resilient thing we can do right now

what's the best inference on policy for our beachside

community not what is the gaussian distribution of tsunami waiting times

do people hopefully see a few directions in which

active inference steps into these issues about uncertainty

here's some of the formalisms from the paper they write our objective is to optimize beliefs

so an approximate posterior over policies π as always

and their consequences which are hidden states s

and those are basically s triple line
it's like a super equal sign
and it's saying s is uh related to this
tau
the t is like the tau the policy horizon
so the hidden states are s from some
initial state
s one stay at time one until the policy
horizon tau so that's the
length of the prediction the depth
through time
um given the observations o from up
until time
until the current time t this
optimization can be cast as minimizing
a generalized free energy functional f
of
so it's a function of something and that
function can be enormously complex like
a neural net
is like a function it's a mapping of
inputs to outputs it's a map
functions are maps and so f of
everything it's like
that's the biggest possible wormhole but
that's why this is a powerful approach
and then it's q of s
and pi so this functional is around
q which is some distribution of states
and policy so
the world state is like what the actual
my estimate
about the world's state is going to be
and the pi is my policy
so here's what is being asked here for
those who might be learning about active
inference or wanting to really
go into these formalisms for perhaps

even the first time
this is like asking how can we have the
best
policy over a given time horizon
so that's the big control theory
question given τ
and given the distribution of
affordances π
what's the best π over τ not
division just what is the best π over
the time horizon τ
so if my policies are left or right um
in a you know
a collision situation and then there's a
time horizon
you only have those two choices and so
you can pull back to the metagame and
say
do i have another choice there's always
space for innovation and for
breaking out of the box and for
reframing it but
at this first base level let's just
focus on
a certain time horizon and a certain set
of policies that are fixed so we're
playing like connect four here or we're
playing like backgammon it's
gonna work for these models too and we
wanna ask
how can we estimate the consequences of
 π
so of our self of the policies of others
and the policies of groups
and the way that we're going to do that
mapping
rather than go infinitely deep on what
is a causal model of the world or have

every atom simulated
in the world we're actually going to go
to a different way and we're going to
ask
how are observed states related to
our estimates of hidden states and how
does that relate to
our underlying causal model of how
hidden states
change through time and how our policy
influences them
that's the framing and there's times
where you can like you know
breathe as much as you want but you're
not going to change by yourself the CO_2
in the atmosphere so by modeling it this
way it doesn't mean that there's going
to be a way
to make the estimate the agent might not
be set up in a way to succeed
but again it's the framing here that
matters and you could frame a winning
chess game or a losing chess game or
a million pawns versus one or any board
size all this
can be included that's what we're using
these numbers and letters for
because then the pie can be really
nuanced it could be
a million different self-driving car
parameter settings
but it could be evaluated they continue
this generalized free energy has two
parts the first entails the generative
model for state transitions
given policies while the second entails
a generative model for policies that
depends on the final states

so emitting constants so the generalized
free energy of sophisticated inference
which is deep time inference
is going to have two parts one part is
going to be a generative model they're
both general models
but they're kind of like two-stroke
engine one generative model
is about state transitions given
policies
and then the other half of the
generative model is about policies
given state estimates of the world
and so those are manifested in
 q and p and they're both
related to policy but they're related in
really specific ways so more details are
provided
above this is just the um
screen screenshot from the paper but
we have two key equations that come at
the bottom here
 q so mind your q 's and p 's just classic
 q is conditioned given policy
it's about observations and states
given policy and it's related to
a model of how observations arise from
states o conditioned on s
and are related to another model about
how
states are conditioned on policies
okay so kind of makes sense because
we're conditioning here
on states i'm sorry we're conditioning
on this right side
 q with policy and then here on the p
we're conditioning on s but then we're
doing an

s estimation here so we like maybe it
kind of cancels that's how we
ultimately get s on the left side here
like this one
somehow scoots to the left side ask a
statistician
but there's more nuanced layouts this is
just the simple one
and um it's basically related this p and
q internally
maybe there's another i don't see how
cute
i don't think it's recursively defined i
think it's two components
of q but let's hear the clarification
from the authors
then there's this p function and that is
related to policy selection itself
so it's not conditioned on policy p
is about policy and this is related to
expectations across policies
and the free energy generalized free
energy of the policy
so p is like the policy selection side
and q is like the inference side
because this is like what are the states
and the observations of the world
given policy that i do given that i go
on a run every day what would be the
states and the observables of the world
that would be different
so it's a counterfactual but i'm framing
it within this type of
approach and then the other half
is given that that's a likely or a
possible or a preferable
or a curious or a fun or a uh
you know habit-forming whatever it

happened to be however it fits into e
and g defined
formally that relates to the policy
selection
path that ends up getting enacted by p
so

yep definitely let's get some more
clarification from
the formalists themselves
they continue with this section equation
1.3

and so here's something pretty
interesting this is
um they write this corresponds to a form
of approximate bayesian inference
i.e variational base in which equation
1.3

is iterated over factors of the mean
field approximation to perform a
coordinate descent or fixed point
iteration

so they are saying that this
formalization corresponds to approximate
bayesian inference i'm going to provide
a little bit more information on that in
a second

i found that to be kind of an
interesting claim and uh
not sure are there mathematicians or
machine learning

experts who could clarify this issue
that would be appreciated

and here's what they wrote which i
thought was really interesting
sophisticated inference recovers bayes
adaptive reinforcement learning in the
zero temperature
limit so cool because it's like saying

in the extreme
no noise case in some info thermodynamic
way
it converges to a certain type of other
thing so just like
some algorithms might converge to a
normal distribution let's just say you
sample a bunch of weights and they
converge to normal
this is like a convergence but it's
converge this model family
is converging on bayes adaptive
reinforcement learning
so we take all that we know about
bayesian adaptive reinforcement learning
all those neural nets
and then now we have the best one and
then it's about converging
into that best funnel
in a higher temperature space that's
even going beyond the model parameters
of those bayesian approaches
quoting the authors again both
approaches sophisticated inference and
the bayes adaptive
reinforcement learning perform belief
state planning
where the agent maximizes an objective
function by taking into account how it
expects its own beliefs to change in the
future and evinces a degree of
sophistication
aka being good at chess being
sophisticated
go the key distinction is that bayes
adaptive reinforcement learning
considers arbitrary reward functions
while sophisticated active inference

optimizes an expected free energy that
can be motivated from first principles
key difference people who are looking
for a key difference
between free energy and something else
that was one
while both the bayesian approach and
free energy can be specified for
particular tasks
the expected free energy additionally
mandates the agent to seek out
information about the world beyond what
is necessary for solving a particular
task
so that's extremely key this goes beyond
for example framing the potential value
of play
in terms of the estimated values of the
deliverables
this actually um something that we've
heard so many participants on the active
stream like blue
and others highlight is like making
space for play
and this full formalization of the
expected free energy
it includes the information gain as well
as the value driven components and some
other components
so actually on equal footing down to
even the units of
information used at least according to
this first interview i
listened to a couple days ago um it's on
our one of the playlists on our channel
um the same natural unit or at least the
same
framework whether there's a coefficient

that balances them he seemed pretty negative on that but i think there could be another way that they're integrated in a more nuanced uh way than just simply an 80 20 20 80. this isn't bipartite partitioning this isn't nature nurture

1.0 this is relational causal networks and so it's going to have a really different way of phrasing it it's not going to be just a coefficient but it'll be like a coefficient i believe because there's systems where it is like it's 80 20 or it's 20 80. in terms of delivering on um something standard versus novelty so like we talked about scripts and 10 you know there's the scripts like the ritual that are meant to be carried out exactly then there's the ones that are unstructured so that whole space back to the authors this model allows inference to account for artificial curiosity that goes beyond reward seeking to the gathering of evidence for the agent's existence i.e its marginal likelihood this is sometimes referred to as self-evidencing and the citation there is the howie 2016. and so we can think back to active inference 8 when we were thinking about that mountain car and about how it just wanted to explore but by exploring

in that admittedly trivial setup
it did discover the goal in the way that
the optimal
um reward uphill climber got ground down
so how can we move beyond just one well
settings move into more nuanced maybe
even semantic innovation landscapes and
optimization landscapes
and converge to approximate bayesian
inference
in the truly formal case where we need
to
but then relax those assumptions with
instrumentalism
and use approximate approximate bayesian
inference
empirically empirically just as
statisticians as scientists as policy
planners
that's really the question in the
doorway all right another
from the same sophisticated paper
amortized inference
effectively and also see active
inference stream eight
with um alec who gave us some really
great information about this and
hopefully we'll have alec on a future
discussion as well
so amortized inference is learning to
infer
variational autoencoders can be regarded
as an instance of amortized inference if
we ignore conditioning on policy
clearly amortization precludes online
inference
that's like real time like data in the
loop and as such may appear biologically

implausible

however it might be the case that
certain brain structures learn to infer
e.g the cerebellum might learn from
inferential processes implemented by the
cerebral cortex

so this is definitely related to a
neurophysiological argument for like
grounding the plausibility
of certain computations being performed
or as if

they were being performed by certain
brain regions that have certain macro
histological arrangement or micro
histological structure

sounds great um i would say
in the case of insects um
could we study whether they're doing it
or not or i just think

maybe it's simpler or it will help us
realize that oh wait we don't really
need the neurons to be
doing that we just need to be able to
think about it just like we didn't need
the neuron to be doing

a linear regression for us to do a
linear regression across neurons

so i don't think that using these
frameworks necessarily
casts the system being analyzed into the
same statistical framework

any more than the linear regression
frame does so that's to say
they all do implicitly and that's the
transparent concept that's the water we
swim in

given that it's the water we swim in and
there's a few different fish tanks

i just don't see how using free energy
 on a system makes you
 strongly committed to that system being
 doing a certain thing
 another part that was interesting from
 this paper was this
 focus on separating the intrinsic and
 the extrinsic value
 and so there's a lot of ways that this
 can be broken down but basically a few
 cool
 equations so this is the divergence the
 kobach libra
 kl informational divergence which is a
 measure of how distant two distributions
 are informationally
 and um the kl divergence between this q
 and the p
 is seen as the risk
 and that is equal to the um like the
 risk of certain states
 like whether the red light is going to
 signal something or not
 and then um the equal sign
 on the other side there's risk in terms
 of the outcomes
 so here's s condition on π_i here's o
 condition on π_i and then here it's p of
 s instead of p of o
 so this is a divergence from the
 observations
 given the outcome so this is like the
 this q is about observations p is about
 observations
 and it's condition on policy so this is
 divergence in states
 divergence in outcomes plus expectations
 this is about q of observations being

conditioned on policy
so we're kind of functionalizing and
conditionally conditioning on a lot
but this is actually about the states
and this is about the states given the
observations and the policy
and how that is bringing us close to
what
the mapping between states and
observations which is the interesting
part
and so there's so much to say here but
they unpack it in terms of the intrinsic
and the extrinsic value so there's
basically
this um separate sections
i would just say not just in the
interest of time but
just in the interest of knowing that
these formalisms really demand to be
unpacked and mapped in a really formal
way
not just discussed in an introductory
lecture so let's save them for another
time
as far as the general and the specific
cases of this
there's a nice figure here with a few
different special cases that descend
into certain uh different sub areas
so if there's no prior this is like an
uninformed or
optimally uninformed bayesian design so
that's the information maximization
principle
that's info gain if we have no ambiguity
like if the
prediction can be completely perfect as

far as the observations are concerned
like all the thermometers are perfect in
your building for example
then you can have basically risk
sensitive or aware policies that's like
cybernetics
as well as informational control and
kind of
occam's principle i'd be really curious
to learn what they meant here
but maybe that means like argument by
simplicity
in the context of systems engineering
could be pretty interesting
then if there's no intrinsic value so if
it's just pure curiosity um search pure
exploration search
then it can it can converge depending on
how you value
intrinsic versus extrinsic to a variety
of different
types of theory and then also there's
this maximum entropy
which i would also like to learn more
about from the authors about
no ambiguity or prior so i'm kind of
curious how these two
no priors and no ambiguity how does that
relate to the no ambiguity of priors and
then is there one that doesn't have any
with no intrinsic value either like what
happens when you have nothing
all right and this one is going to come
back in a very very similar
format in the
figures for the paper but basically it
starts off
and it's like here's us at the top at

observation time point three
and then there's one two three four
things can happen
and then from thing one or three
happening four things could
be observed and then from one or three
of those or from two or four
then you can have one two three four
things happen
and here outcome one and two and four
lead to one of four observations and so
this
is like an unrolling this is like a roll
out of a deep time prediction
on a tree and that's related to a couple
different ideas which is why we
introduce them
which is the prediction through
recursion
and the counterfactual branching so the
first thing to remember is that the
branches are literally counterfactuals
it's like
what would my prediction be but it's
really just predicting the whole
distribution
so instead of thinking as predicting is
my team going to win or not just think
what is the distribution of likelihoods
of each team winning the world series
and then that's like what is the
likelihood of each of these outcomes
conditioned on this
and you keep on conditioning on
conditioning on and rolling that out
through deep time
so it relates to counterfactuals and
the state

outcome mapping so state of the world
and then mapped onto your observations
the sensory outcomes through this
learning of the relationship
so here is what it looks like in the
format that we looked at at the end of
10.2

you have the transition matrix it's
like there's a state
it emits observation it's this dark kind
then there's an action policy
whether from the agent or from and or
from the world and that
modifies or not the state transition
matrix and then the state
goes to the next state and then through
a which can be similar or not
it emits another observation that could
be probabilistic or not
so we're going to see this again in just
a few slides but

pretty interesting and that's just the
first approach and that shows that the
sophisticated inference paper was the
one that really brought it out
and then also this is pretty interesting
figure eight and here's the caption
this schematic summarizes the various
imperatives implied by minimizing a free
energy functional of posterior beliefs
about policies

ensuring states and subsequent outcomes
the information diagrams in the upper
panels represent the entropy of three
variables

where intersections correspond to shared
information or mutual information
so professor jim crutchfield at uc davis

or anyone

in your area i would love to be

learning more about this question of the
overlapping venn diagrams because i
remember

a lecture with some overlapping venn
diagrams and information theory from
professor crutchfield that was very
influential for me

and then the caption continues a
conditional entropy corresponds to an
area that precludes the variable upon
which the entropy is conditioned

note that because there is no note that
there is no overlap between policies and
outcomes that is outside hidden states
this is because hidden states form a
markov blanket i.e informational
bottleneck between policies and outcomes
two complementary two complementary
formulations of minimizing expected free
energy are shown on the right in terms
of risk

and ambiguity

and left in terms of information gain in
entropy

respectively one can see that both will
tend to increase the overlap

of mutual information between hidden
states and outputs that's what we really
want

is we want the states and the outputs we
want to be

learning that relationship

both will tend to increase the overlap
of mutual information between hidden
states and outputs while minimizing
entropy or bayesian risk

in these diagrams we have assumed steady state such that risk becomes the mutual information between policies and hidden states

for simplicity we have omitted dependencies on initial observations fair the various schemes or formulations considered in the text are shown on the bottom

these demonstrated the bayesian control theory iekl control and bayesian risk and optimal bayesian denying an optimal beijing design figures complementary imperatives

okay bayesian design of what the answer is of experiments so this is optimal experimentation conditioned on everything you know okay so this is like

given the state of what's known how can you get information gained and then here is related to making uh the lowest possible risk through policy given to the states that are known

and that's related to what's known as kl control

and so what we're seeing here is in this mapping of active inference and not necessarily other frameworks

i'd be open to correction or expansion on this but not necessarily in other frameworks

are there um such even ways to understand and equivocate a really broad

range of strategies for example you

could do

really this is kind of like explore and

this is kind of like exploit

except instead of the parameter

switching explore to exploit

or one mode to the other it's actually

like a more nuanced trade-off

mediated through policy and

what that policy is enacting in is

a co-optimization of

information gain that's this outward and

it's expansive in this variational

principle with the downward

coordinating bayesian risk precision

optimization

so the bayesian design is like you're on

this point on this uh

you know wherever you're at and it's

expanding outwards with the info gain

with the optimal experimentation and

then it's collapsing inwards

in order to decrease surprise about the

world

all right now on

april 6th and 13th 2021

casper hesp and other authors from this

recently published paper will be joining

the active stream

so we'll go through the paper and all

the figures and everything

then but

the figures will just go through it one

first time because they're going to be

basically exactly copied over

for the sophisticated affect of

inference so

where we start is m1 and we'll go

through these pretty fast

m1 the first layer of the model is about how hidden states instantaneously are associated with observations so here's observations here's a likelihood mapping here's the states we're estimating states given our prior d and then from observations through the likelihood mapping a some math now m2 is a generative model of anticipation so here we have d our initial priors and then we have b which is how d changes basically how those those priors change and then we have states through time one two three and then at each time we're having either a prediction or a confirmation or updating of an observation that's the time specific predictive density so m2 is like how does m1 which is at one time point change over two or more time points so that's m2 m3 is a generative model of action so this is where we introduce the action layer and so there's a couple of pieces that play into it the first thing to notice is that we have this exact same m2 it's literally it's um the exact same image copied out here's m2 looks kind of like three letters there's three prongs pointing down and a little head with the d same exact shape here okay so m2 the box is clarifying this is

the exact same

m_2 that we just talked about m_3 is

around that

and so how does m_3 come into play

well it's all about π policy and so

π policy plays into the b

matrix which is how the state

estimate changes through time so our

policy is really if I exercise every day

then I'll be getting healthier in this

way so something like that

and the π is again only playing into

these b matrices it's not playing into

the s

state estimation it's not playing into

any a matrix learning it's a different

part of the brain different part of the

model

or so they say maybe and

the π reaches downwards into the b

but also it's getting influenced by a

couple of new things up here

which is the baseline prior over

policies

and the phenotypic outcome preferences

which are c so c

is your preference matrix e is like your

affordances

that's like your your baseline

affordances of your prior of affordances

just like you're endowed from evolution

and from learning and development what

are the tools

that are at the disposal and then

there's the phenotypic preferences which

are also evolutionary priors that can be

updated

and they can be contextual and through g

there's a policy selection okay more
math or math but
it's about d this internal model of how
states and observations are linked and
then we're going to hook
p pi policy in here
through using this g free energy
optimizing function then
g is the interface for
where this generative model of implicit
metacognition comes into play
and so this estimation of meta
confidence so remember we talked about
well it's ten percent that my team is
going to win the world series but i'm
not too confident about
that estimate and so implicitly about
the past
present and future we speak about it in
different ways but we have not just a
state estimate about the world
but a precision estimate like it was
around 70 degrees or
it was exactly 70 degrees i looked at it
on a thermometer
and so here gamma is this
precision parameter and um
beta is the prior over precision
so uh it it's because
beta that is the parameter i think of
this gamma distribution
so the gamma parameter is what is kind
of the high to low temperature
on g so it is like the explore exploit
trade-off
in a sense but it relates in a little
bit more
nuanced and and surely with a lot of

different consequences

um it relates to this um

optimal experimentation versus precision
management

convergence divergence uh

expansive expansive retraction and
tensile and compressive

tensegrity synergetics it relates to a
lot of these other areas

all right so to conclude that section

yeah this one is a little long but it's

fun and it was a fun one to learn about

so um hopefully we can return to this
slide now

and say okay the affect um

and the temporal depth are two pillars

and now we're going to build on top of
them we're going to think about an agent
that goes deep

and one of its predictions is about its
own affect

so here's the figure from the paper that

we're reading from the hesp at all

and the reason why i went through it in
the previous section

was because as you'll see it's very
similar

to the previous version so we have
states and the a likelihood mapping
states and observations of the world
then

we have b the matrix of how states
change in the world

and then we have u which is action here

and u is as before interfacing only with
 b

okay so that's where we've seen before
but now we're going to be estimating

precision

and free energy at each future time

point

so we're going to be considering our

action and our

importantly our precision parameter

which is required

to calculate the g from the top we're

going to have

basically this initial condition d and

then there's going to be a state

that carries over into the next

situation so the next branch down the

row this

this whole thing is gonna get copied

over so you can have these big you know

graphs all these recursive

functions calling each other basically

but this is the state

and then a here u_m is

going to be initially just giving this

first state but

again we're going to be at each time

point rolling out

in a deep sophisticated way task

specific sophisticated inference as in

first in 2020

we're going to be doing this counter

factual analysis

about this deep branching and our

precision

and our valence about it at each time

now here's the illustration of the state

space of the task with the four states

so it starts off neutral and then

basically there's a transition

that's possible so you can imagine a 4×4

matrix with the state transition

probabilities
and the transition matrix that they set
up
has a ability to go from a neutral
to a precise small gain that's two
it's a good probability of getting um a
good outcome
or you can go to this dangerous high
gain
where it's mapped here i believe it has
um
the three so if you land on a one you
start in the stay in the neutral
position
if you get to a two
you can go to either one or three
or they have it connected to two but i
don't know if they mean that edge
um properly then if you get to three
you can hop to one two three or four
again i don't know if they have an edge
there
as well and then if you get to four it's
an extremely
painful state and it's absorbing it only
leads to itself
and so here it's like a dead end so at
the first time point let's just say
we're starting at t equals one
and we're just evaluating i'm in state
one what could happen
okay so here you could go to four now it
you can't on that first one so maybe the
likelihood of
outcome four is zero so you're like wow
all my outcomes are great
i could either have this two or this
three not that those are the values but

you can at least say at least one
road uh one down the road gain of one
gain of two
one step down the road both my outcomes
are good
then you look one level deeper and all
of a sudden
there's a few things happening first is
this branch
is staying very high positive all the
estimates are very positive
but on this branch you're starting to
see that it's possible to get this
negative valence
so now your estimates out of these four
you're like well it could be
pretty good pretty good very good or
kind of negative
so now at each time point not only are
you tracking
the likelihood of different estimates
but really you're tracking it as like a
distribution
and the distribution is about state
estimates and uncertainty and
the variance the prior beta gamma
those are the key pieces here that's why
this is so interesting
and so as you can see from this
graphical
layout this kind of markov state
transition matrix
layout which we could see maybe in the
code
you can see that eventually you're going
to get to four even if only one percent
of the time from three to four
you go to four if you stay in four

you're always going to stay in four it's
an absorbing state
and so eventually if you had 100 little
ants or 100 particles and they started
here they'd all end up in four
if you set up that way and they can't
escape and so
here's what we end up seeing well first
before we get we see these are just the
mathematics
we're not going to go into it in this
one but they're described there and
they're described here and here
and here's what you get when you do the
simulation and so remember that the
uh negative states were absorbing and
there was just
an increasing number of them and so four
charts all on this time scale from zero
to two thousand
so the first thing to notice is that the
perception
of the good versus bad it starts in the
middle
then it goes high maybe as it's
discovering some of these shallow
branches where there's a lot of winds
but then around 500 it starts to really
slip down and that's in terms of how
many states it's um
playing out the simulation in okay
and now so even though the simulation
each round is totally
winnable let's just say if there was any
agency in the situation
by imagining that there's an infinite or
a growing number of future outcomes that
are negative

it leads one to have bad affect in the way that they modeled it in this model another interesting feature is that initially as evidence starts pouring in precision increases but then around the same time that the valence drops the precision drops basically as well this looks like a crypto just dying but the precision dropping here going from

a high of around 1.8 to florian at 0.6 that is uh showing that there's a increased variance of the estimate it's like would you rather know that you're going to have ten dollars plus or minus one

or ten dollars plus or minus 40. it's like wait i might

have negative 30 from this deal that sounds like very different maybe i'll just take 10 plus or minus 2.

and so that's related to risk tolerance and design of experiments

okay let's look at this bottom right and then we'll go to the top right

the bottom right is the threat perception which is the fraction of imagined events that are negative

so again at one temporal depth there's zero percent of these uh outcomes are negative

because we haven't gotten there yet in this first step

none of them can be negative you can't get any of the bad outcome

but as more and more simulations roll out

there starts to be this converging

amount that are negative
now here again they just called it a
proof of concept but
they're just showing one parameter
combination so it's not a statement
about
how a certain natural system is in just
any way it's a framework for thinking
about it but as
the threat perception increases and
levels off
there's also a stabilization of a
uncertain predictive posterior for
action
so when things are good precision is
high
and the posterior on action is obvious
so it's like very clear we're getting a
lot of big winning states and there's a
very low posterior for a lot of negative
states
then as the percentage of threats starts
to increase
and interestingly it's actually after it
starts to get a little
better but maybe the state space is just
branching out even further
as that happens there's a phase
transition in the precision and the
valence which in active inference are
very tied up because anxiety
which is a negative emotion we're saying
in this one in excess
is increased by uncertainty and so
basically this is like a good level of
uncertainty
our posterior action was pretty we're
pretty clear about which way we were

going to be able to go
but then this is like even though the
green is still
at sixty percent um the teal is kind of
at like
uh you know so the good ones are not
losing
most of it is actually rewarding but
there's still a small fraction
that are damaging but it turns out that
by having that level of uncertainty
it generates negative affect negative
valids through this low precision of the
model
so that's pretty interesting in this
reduced action model precision is
interesting too
so that's basically the figures of the
paper
it was relatively long but again the
paper was short
so i hope that you read it but at the
same time it clearly builds so directly
upon the other
two work that it just didn't make sense
to only read the formalisms from this
paper
when it was so linked to previous work
all right so the concluding questions
are just
what are frameworks for affect
and anticipation what would a good
framework for affect or
anticipation enable for us what does
affect encompass other than valence like
other than the plus minus
what other features of affect are there
what are some unique predictions and

some experiments
of this model whether in general or
in the specific situation but like what
is something that this model predicts
that we might be able to go out and test
or measure or already have the data to
know
and then what are the next steps for
active inference
how will active inference
increase to another level of
sophistication or include another group
of researchers in the conversation
or integrate best or emerging practices
from another domain
how can active inference grow from this
encounter
and serve as a you know helpful
reference point for the other fields as
well
and then what is the goal of this
research
um they were mapping things onto anxiety
but of course it's just a parameter and
a model
and certainly few authors would say that
their models are anything but parameters
and models
yet it's so difficult to especially the
multiple senses of anticipatory systems
and living systems and affect in
different systems robotic systems
it's at the nexus where it's almost like
granting
agency to models but then removing
agencies from humans
so you know having affect in a model but
then not

caring about a human affect when they're
telling you so that's sort of a
not not of course saying that the
authors are doing that or part of that
or anything like that
it's just that when we have these
affective emotive agents it's like
yeah that's us that's how we should be
respecting people
so how can we build on that
and take this model which seems to be
really affirmative
about human agency and choice at the
very least enabling of human agency
in a way that other models again not
that they weren't
but that is different that's kind of the
discussion that we
are having and that's what's been so
awesome so
thanks for participating everyone it was
kind of a fun
discussion a little bit more than an
hour and a half
we will provide follow-up forms to the
live participants
so anyone's feedback suggestions
questions would be
super helpful and awesome stay
in communication with us and uh
yeah we hope to see you in these last
two discussions of 2020
and then the new 7 a.m to 9 a.m pst
2021 i'm really looking forward to that
i know a lot of our teammates are as
well
so check it out go to the website
get the updated information thanks

everyone for

listening in live or replay

and it was a fun discussion i hope that

you

got something out of it alright thanks

and talk to you later